

Evidence-Centered Visualization for ML Security Engineering

Aggregating Repository-Level Measurements Into Traceable Technical Dashboards

CONTRIBUTION

An evidence-centered dashboard server that aggregates committed per-repo JSON into traceable HTML + a token-authed REST API, deliberately surfacing weak results (e.g. ROC-AUC 0.54, OOD F1 0.72) alongside strong ones rather than curating highlights.

METHOD

BOUNDARIES / LIMITATIONS

- Static evidence snapshots, not live monitoring.
- Data can lag the current CI state of each repo.
- No alerting or real-time telemetry.
- Depends on sibling repos committing evidence.
- Local aggregation; not a hardened service.

CURRENT VERIFIED EVIDENCE

FastAPI	Dashboard framework
token	Auth on API
shared	Design system
97	Grouped F1 (redteam)
72	OOD F1 (redteam)

EVIDENCE NOTES

54	Weak ROC-AUC (poison)	• VERIFIED_AT_SNAPSHOT - Aggregates evidence from 8 active ML security repos - README; dashboard_server.py discovers sibling evidence/*_json.
8	REPOS AGGREGATED	• VERIFIED_AT_SNAPSHOT - 37 test functions - Counted def test_ definitions in tests/ at source snapshot b02a0fc; current README also records 37 passing pytest...
37	TEST FUNCTIONS	• VERIFIED_AT_SNAPSHOT - Shows weak results explicitly (OOD F1 0.72, ROC-AUC 0.54) - README states weak-alongside-strong design; numbers sourced from sibling...

REPRODUCE

```
python dashboard_server.py
pytest tests/
```